

Sai Karan Mekala

Software Engineer

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SUMMARY

Software Engineer with 5+ years of experience building scalable full-stack and frontend-heavy applications with React.js, TypeScript, Java, Spring Boot, and microservices. Strong focus on UI engineering, AI-powered automation, cloud-native systems, and distributed backend architecture. Experienced in designing highly responsive frontends, developing secure backend APIs, and deploying cloud applications using AWS. Skilled in CI/CD, containerization, system design, data modeling, and performance optimization. Adept at Agile development and cross-functional collaboration to deliver modern, high-quality engineering solutions aligned with U.S. software engineering expectations.

TECHNICAL SKILLS

Programming Languages: Java, Python, JavaScript (ES6+), TypeScript, SQL, CSS3.

Frontend: React.js, Next.js, TypeScript, Redux, HTML5, CSS3, TailwindCSS, SPA, Component Architecture, UI/UX.

Backend: Java, Spring Boot, Node.js (Express.js), Hibernate, JPA, Python (API Development)

Cloud: AWS (EC2, Lambda, RDS, S3, CloudWatch), Azure (Basics), GCP (Familiar), Serverless, Load Balancing

DevOps & CI/CD: Docker, Kubernetes, Jenkins, Git, GitHub Actions, Terraform, SonarQube, CI/CD Pipelines, Monitoring & Observability

Databases: Snowflake, Redshift, PostgreSQL, MySQL, MongoDB

Development & Methodologies: Agile, Scrum, Kanban, Test-Driven Development (TDD), SDLC, Behavior-Driven Development (BDD), Code Review Practices, Software Design Patterns

AI & Automation: OpenAI APIs, LangChain, AI-assisted automation, Python scripting, Workflow Automation

Microservices & Architecture: REST APIs, Microservices, MVC, Event-Driven Architecture, Kafka, RabbitMQ, GraphQL, System Design, Performance Optimization.

PROFESSIONAL EXPERIENCE

Software Engineer

Fiserv, USA | Aug 2024 – Present

- Engineered scalable payment microservices using Java, Spring Boot, and event-driven architecture, processing 1M+ daily transactions with high reliability.
- Developed React.js + TypeScript dashboards for real-time transaction monitoring and fraud analytics, improving operational visibility by 45%.
- Implemented AI-driven workflow automation using Python and OpenAI APIs, reducing manual review and support operations by 35%.
- Enhanced platform security with OAuth2, JWT, and TLS, achieving 100% regulatory compliance and significantly reducing risk exposure.
- Automated CI/CD processes using Jenkins, Docker, Kubernetes, accelerating deployment cycles by 40% while maintaining 99.9% uptime.
- Optimized PostgreSQL queries and caching strategies to boost backend throughput by 35%, supporting 500K+ concurrent users.
- Deployed high-availability systems on AWS (EC2, Lambda, RDS, S3) using IaC and load balancing, achieving 99.95% service reliability.

Software Engineer

Persistent Systems, India | Jan 2022 – Aug 2023

- Designed and maintained Spring Boot and Hibernate microservices integrated with PostgreSQL for enterprise application modernization.
- Refactored and optimized REST APIs, improving service performance and response time by 25% across multiple product lines.
- Built AI-assisted testing and debugging utilities using Python, reducing regression testing efforts by 30%.
- Enforced strong code quality through JUnit, Mockito, and Postman testing frameworks, significantly reducing production defects.
- Streamlined DevOps workflows via Jenkins, Git, Docker, accelerating release cycles by 50% through automation.
- Partnered with product, QA, UX, and business teams in Agile sprints to deliver features with 95% on-time completion across 5+ major initiatives.
- Implemented Agile and Scrum development methodologies, following best practices such as TDD and code reviews to ensure scalable, maintainable, and high-quality software delivery.

Full Stack Developer

Cybage Software, India | Oct 2019 – Dec 2021

- Developed user-focused full-stack applications using React.js, Node.js, and MySQL, supporting 100K+ enterprise users.
- Delivered responsive UI components using React.js, HTML5, CSS3, and SPA patterns, improving user experience by 40%
- Integrated secure REST APIs utilizing JWT authentication, reducing security vulnerabilities across 50+ client deployments
- Automated deployments on AWS EC2 with CI/CD pipelines, reducing release times by 45% and improving deployment consistency.
- Collaborated with QA and UX teams to resolve 200+ performance and UI issues, improving usability scores by 30%.

PROJECT

- **FinFlow – Cloud-Based Digital Payment Platform**

Technologies: Java, Spring Boot, Spring Cloud, PostgreSQL, AWS, Docker, Kubernetes, Jenkins, OAuth2, JWT.

- Designed backend microservices for payment and user modules, managing 10K+ daily transactions with optimized schema and database design.
- Built secure RESTful APIs with OAuth2, JWT, and Spring Security, ensuring 100% authorization compliance for 5K+ users.
- Integrated payment gateways and banking APIs with end-to-end testing, enabling real-time processing with 99% uptime and zero transaction failures.
- Deployed services on AWS (EC2, RDS, S3) using Docker and Kubernetes, scaling to handle 1M+ requests per hour.
- Optimized Jenkins CI pipelines and implemented CloudWatch monitoring and logging, reducing deployment errors by 40% and improving system reliability.

EDUCATION

Master of Science in Information Technology and Management

Concordia University, St. Paul, MN, USA - Aug 2023 – Apr 2025

Bachelor of Technology in Information Technology

Teegala Krishna Reddy Engineering College, India - Graduated 2022